

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: ARTIFICIAL INTELLIGENCE

AIM: Breadth First Search & Iterative Depth First Search in Python.

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

OUTPUT:



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IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/T095/AI PRAC 1.py =====
BFS: Lokhandwala -> Oshiwara Link Road -> Oshiwara -> SV Road -> Juhu Circle -> Juhu
Steps: 5 | Nodes visited: 14 | Time: 5.01000004078378e-05

DFS: Lokhandwala -> Veera Desai Road -> Andheri Link Road -> JP Road -> DN Nagar -> Juhu
Steps: 5 | Nodes visited: 6 | Time: 2.249999943160219e-05

Comparison
Shorter path: Equal
Fewer nodes visited: DFS

Time Complexity: O(V + E) for both
Space Complexity: O(V) for both

>>>
```

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